

IEEE 1474 vs IEC 62290

The two standards a US CBTC specification has to satisfy at once — what each one actually governs, and which clause to cite for what.

ASPECT	IEEE 1474	IEC 62290
Origin	North America; driven by APTA and transit agencies.	International; driven by IEC working groups and European operators.
Scope	System performance, availability, headway, braking, communication latency.	Functional requirements, architecture, safety standards, GoA classification.
GoA discussion	Minimal; assumes operator-driven systems.	Central; defines automation levels GoA 0–4.
Safety specification	Implies SIL 4 but is less prescriptive.	Explicit SIL 4 for ATP; SIL 1–2 for ATO/ATS.
Procurement focus	Detailed performance metrics, availability targets, spare parts.	Functional architecture, automation level, operational concepts.
Use in US projects	Cited in procurement RFPs; directly constrains vendor performance.	Increasingly cited for GoA classification and automation expectations.

The distinction that settles most specification arguments. IEEE 1474 says “the system must achieve 5.5-minute headways with 99.5 percent availability.” IEC 62290 says “the system must execute ATP at SIL 4 and support GoA 2 (semi-automatic operation).” A specification that cites only the first has not constrained the architecture; one that cites only the second has not constrained performance. Current US best practice cites both.

Where each belongs in an RFP

RFP SECTION	CITE	BECAUSE
Performance & availability	IEEE 1474	Headway, availability targets, latency budgets and spares are quantified here and are directly enforceable against the vendor.
Functional architecture	IEC 62290	The ATP / ATO / ATS decomposition gives design review a template for verifying that safety-critical and non-critical logic are actually separated.
Operational concept	IEC 62290	A GoA level states operational intent unambiguously — “GoA 3 with on-train staff at peak, unattended off-peak” — where “driverless” does not.
Safety case	Both	IEC 62290 fixes the SIL allocation; IEEE 1474 fixes the availability and performance the safety argument has to hold under.
International bidders	Both	Referencing IEC 62290 lets European and Asian vendor proposals be evaluated against consistent criteria instead of translated ones.

Watch the living standard. IEC 62290-3 and later parts cover cybersecurity, onboard communication protocols and test specifications. Amendments from 2020–2025 expanded driverless-metro operation, energy efficiency and cybersecurity resilience — the last of which matters more every year as agencies move ATS platforms to the cloud.